

1. Administrative & Study Identification

Full Study Title: A Post-Authorization, Long-term Study of Ozanimod Real-world Safety **Short Title/Acronym:** ORION **Protocol Number:** BMS-ORION-PASS **NCT Number:** NCT05605782 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** Ozanimod (Zeposia) **Phase of Development:** Phase IV (Post-Authorization Safety Study / Observational) **Medical Monitor:** Bristol-Myers Squibb Medical Affairs Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To determine the rates of adverse events of interest (AEIs) in a real-world population of participants with relapsing-remitting multiple sclerosis (RRMS) receiving ozanimod, compared to non-exposed patients receiving other therapies. **Secondary Objectives:** To evaluate real-world utilization, treatment discontinuation, and the comparative long-term safety profile of ozanimod versus other S1P receptor modulators and non-S1P disease-modifying treatments (DMTs).

2.2 Background & Rationale To address the gap in long-term safety data in a real-world setting, this study is being conducted as a Post-Authorization Safety Study (PASS). It monitors the safety of ozanimod—a sphingosine 1-phosphate (S1P) receptor modulator—over an 11-year follow-up period to ensure ongoing patient safety and fulfill regulatory post-market commitments for RRMS treatment.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Multinational Registry / Database Cohort Study) **Control Type:** Active Comparator (Other S1P modulators and non-S1P DMTs cohorts) **Blinding:** Open-label (Real-world observational) **Randomization:** N/A (Observational assignment based on actual clinical prescriptions)

3.2 Planned Enrollment & Duration **Target N\$:** ~9,000 participants **Number of Sites:** Multinational (Utilizing various regional healthcare data sources and registries) **Estimated Study Start:** October 2022 **Estimated Primary Completion:** July 2033

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Men and women age \$ge 18\$ years old. 2. Have a diagnosis of multiple sclerosis (MS) recorded on or

before the index prescription date. 3. New users of ("initiate") treatment with one of the three cohort-defining treatments (Ozanimod, other S1P receptor modulator, or a non-S1P DMT). 4. Have at least 6 months of continuous enrollment in the data source (providing medical and dispensing/prescription history data) before the index date.

4.2 Exclusion Criteria 1. Prior exposure to ozanimod before the index date (for the comparative cohorts). 2. Missing or incomplete baseline health records for the 6 months prior to the index prescription. 3. Concurrent participation in interventional clinical trials with investigational drugs that would confound real-world safety assessments.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ozanimod standard oral daily dose (typically 0.92 mg following initial titration), as prescribed in real-world clinical practice. **Route of Administration:** Oral **Duration of Treatment:** Long-term exposure spanning up to 11 years of observational follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care disease-modifying therapies and concomitant symptom management medications per routine clinical practice. **Prohibited:** Use of unapproved investigational agents that may interfere with real-world registry claims.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-6 to 0	Screening / Baseline
2	0 to 11	Index Date (-6 months to 0) Identification of index prescription, extraction of 6-month continuous enrollment medical history Interim Continuous Ongoing electronic health record/claims extraction for AEIs, treatment switching, and discontinuation End of Study Up to Year 11 Final claims extraction, statistical adjustment, and overall safety data aggregation

(Note: As a registry-based observational study, traditional scheduled onsite visits are replaced by continuous data source monitoring.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence rates of targeted Adverse Events of Interest (AEIs) during the observation period. **Measurement Method:** Extraction and longitudinal analysis of diagnostic and procedural codes from real-world data (electronic health records, claims data, registries) over 11 years.

7.2 Secondary & Exploratory Endpoints 1. Rates of treatment discontinuation and switching. 2. Comparative risk profiles of AEs between the ozanimod cohort and the other S1P receptor modulator/non-S1P DMT cohorts.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via continuous algorithmic scanning and extraction of healthcare databases/registries for predefined safety signals. **Serious Adverse Events (SAEs):** Captured through hospitalization records and severe diagnostic codes within the real-world data set. **Data Monitoring Committee (DMC):** External Pharmacoepidemiology and Safety Review Board overseeing the PASS progress.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all matched patients newly initiating one of the cohort-defining treatments with ≥ 6 months baseline data). **Sample Size Justification:** $N \approx 9,000$ to provide sufficient statistical power for detecting rare or long-term adverse events over an 11-year follow-up period. **Handling of Missing Data:** Propensity score matching/weighting to adjust for baseline demographic and clinical covariates; imputation applied based on standard pharmacoepidemiologic guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local regulatory mandates for post-authorization safety studies. **Monitoring Plan:** Periodic safety update reports (PSURs) submitted to regulatory agencies (e.g., interim reports every 3 years). **Audit Trail:** Secure maintenance of anonymized patient records, with rigorous validation of diagnostic and claims code extraction.

11. Study Identifiers & Governance

Primary ID: ORION **Registry Number:** NCT05605782 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** ORION **Official Regulatory Title:** A Post-Authorization, Long-term Study of Ozanimod Real-world Safety

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing Remitting Multiple Sclerosis (RRMS)
Diagnosis Threshold: Confirmed diagnosis of MS on or before index prescription
Medical Methodology: Disease-Modifying Treatment (DMT)
Optimization Target: Long-term Safety and Tolerability tracking

13. Study Procedures & Methodology

Management Pathway: Routine real-world clinical practice
Education Protocol (HCP): Standard label instructions for Ozanimod
Education Protocol (Patient): Standard patient counseling per real-world practice
Quality Audit Mechanism: Healthcare claims and registry database validation

14. Observation & Follow-Up Schedule

Total Duration: 11 Years (Up to July 2033)
Onsite Visit Cadence: N/A (Real-world data extraction)
Remote Monitoring Frequency: Continuous registry database extraction
Checklist Review Interval: Annual and Triennial data cut/analysis for safety reporting

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				